

4.10 Antiderivatives Worksheet

1. Find the most general antiderivative for each of the following. Use algebra to help simplify before trying to find antiderivatives when appropriate.

a) $f(x) = 10x^4 - 2\sec^2(x)$

b) $f(x) = \frac{x+2}{x} + x^{3/2}$

c) $f(x) = 4\sin(x) - 2x^3 + \frac{\cos(x)}{2}$

d) $f(x) = \frac{-5}{\sqrt{1-x^2}} + \frac{4}{\sqrt{x}}$

e) $f(x) = \frac{x^2 - 16x^3e^x + 8}{8x^3}$

2. Find $g(t)$ given that $g''(t) = 3t^3 + 3\cos(t)$, $g'(0) = 7$ & $g(0) = 10$.

3. The acceleration due to gravity near the surface of some planet is $-4m/s^2$. An object is shot upward from the surface of this planet and 4 seconds later it has fallen back to the surface. What is the initial velocity of this object (be sure to include units in your answer)?

4. A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6 \text{ m/s}$ and its initial displacement is $s(0) = 9 \text{ m}$. Find its position function $s(t)$.
5. Find f if $f''(x) = 12x^2 + 6x - 4$, $f(0) = 4$, & $f(1) = 1$.
6. A ball is thrown upward with a speed of 48 ft/s from the edge of a cliff 432 ft above the ground. Find its height above the ground t seconds later. When does it reach its maximum height? When does it hit the ground? (Acceleration due to gravity is -32 ft/s^2)